

Supply Chain Analytics Project

Executive Summary

Project Overview

This project analyzes supply chain performance using PostgreSQL for SQL analysis and Power BI for interactive dashboards. The goal is to identify operational inefficiencies and generate actionable business insights.

Business Objective

Evaluate product performance, supplier efficiency, inventory levels, logistics, manufacturing costs, and quality metrics to support data-driven decision-making.

Tools & Technologies

PostgreSQL, SQL, Power BI, DAX, Microsoft Excel.

Key Findings

- Skincare products generated the highest revenue and sales volume.
- Revenue was concentrated among a few suppliers.
- Some products showed high demand despite lower inventory levels.
- Shipping performance varied across carriers.
- Manufacturing costs and defect rates differed by supplier and product type.

Business Recommendations

- Improve inventory planning for high-demand products.
- Strengthen relationships with high-performing suppliers.
- Reduce defect rates through supplier quality reviews.
- Optimize shipping carriers based on delivery performance.
- Monitor manufacturing costs to improve profitability.

Dashboard Overview

The Power BI report includes two pages: Executive Overview and Operations & Quality, with interactive KPIs, slicers, and business visualizations.

Conclusion

This project demonstrates an end-to-end analytics workflow using SQL and Power BI to transform operational data into actionable business insights.